

Fig. 4: Electron micrograph of nanoantenna

assembled. Further, two technological parameters are to be fulfilled. First, the antennae must be close to the size of the wavelength being captured, which in the case of the solar spectrum can be very small—from millimetres down to a few hundred nanometres. Second, the currents produced will be alternating at frequencies too high to be of use unless first converted into a steady current. The problem here is that silicon diodes, which are crucial to the conversion, typically don't operate at the high frequencies required.

Both of these barriers are now broken. A technique for creating arrays of billions of antennae that are small enough to work with mid, near infrared and far ends of the infrared spectrum is developed. Current-conversion problem is tackled by creating novel diodes capable of handling high optical frequencies.

Semiconductor diodes act like valves, converting alternating current (AC) into direct current. To work with the new-generation antennae, these have to operate at the AC frequencies being received and match the conductive properties of the antenna. Semiconductors are ill-suited for this, as these tend to become less conductive when shrunk to the size of the antennae.

It is found that physical geometry of the device influences current flow. By creating asymmetry in the geometry of the diodes, it funnels electrons to flow only in one way. Nanoantennae are optical collection devices designed to absorb specific wavelengths that are proportional to the size and shape of the nanoantenna. Just like their bigger

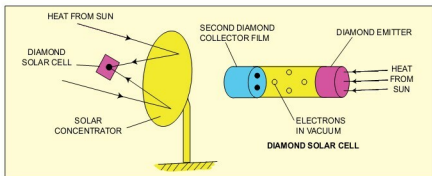


Fig. 5: Solar diamond thermoelectric generator

radio frequency counterparts, these have the advantage of limiting the incoming radiation to one mode with one polarisation.

Several different types of nanoantennae—such as dipole, bow-tie, spiral and log-periodic or microstrip antennae—are used to generate power from optical radiation to suit the specific wavelength. These can be used in the visible and invisible ranges. A nanoantenna is fabricated using electron-beam lithography and lift-off technique.

Typically, a dipole antenna element is made of 100nm-thick evaporated gold. A bolometer made of nickel is patterned at the feed end of the nanoantenna to serve as the detection mechanism. A pair of 200μm bond pads is patterned on both ends of the dipole antenna in order to provide biasing and obtain electricity. The device is fabricated on a silicon wafer with 200nm thermally grown silicon dioxide. An electron micrograph of the device is shown in Fig. 4.

Diamond power

Diamond no bigger than a speck of dust might not interest a bank robber, but is turning out to be a physicist's best friend. Diamond's hardness and lustre have been prized for centuries, but when it is reduced to nanoscale size, other valuable properties of electronics emerge. Diamond films can make a new kind of solar cell that generates electricity by absorbing heat rather than visible light wavelength. The property is known as thermionic emission—the property of some materials to spit out electrons when

heated. It turns out that ultra-thin diamond is better at this than most other materials.

The device is made of two thin films of diamond a few micrometres thick, which are separated by a vacuum a few hundred micrometres thick. A reflective dish is used to focus sunlight onto the device. As sunlight heats the outer diamond film (the emitter), the hottest most energetic electrons fly off. These travel across vacuum to a second diamond film (known as collector) that generates current.

A bank of such diamond-based solar concentrators can be made to exploit the sun's energy to electric power. To make the diamond films work effectively, lithium atoms must be implanted into them. These atoms form positive charges near the film surface, which helps hot electrons leave.

Unfortunately, the very arrangement of carbon atoms that gives diamond its hardness makes it difficult to insert alien atoms. Lithium atoms seep extremely slowly in the hot diamond film. Hence lithium ions are used, which diffuse more easily throughout the structure.

The arrangement of a solar diamond thermoelectric generator device is shown in Fig. 5. A number of such devices arranged in arrays can be utilised to harvest the sun's thermal energy in the form of electrical energy. The efficiency of this system is much more than the conventional solar thermal power generator. ●

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